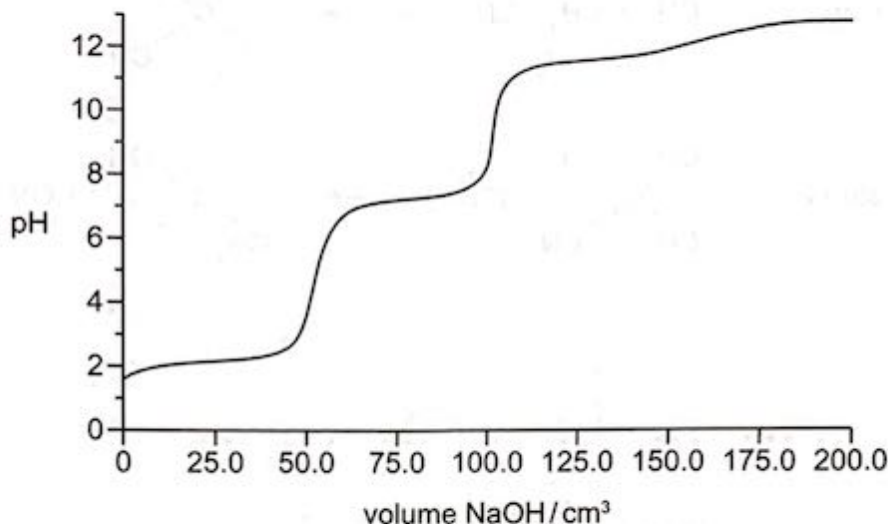


- 15 When an acid is titrated with a strong base, an equivalence point is reached when the amount of base added is equal to the amount of H^+ dissociated. The curve for the titration of 50.0 cm^3 of 0.10 mol dm^{-3} phosphoric acid, H_3PO_4 , a triprotic acid, against 0.10 mol dm^{-3} NaOH is shown.



The table shows the pH range of colour changes of some acid–base indicators.

indicator	pH range of colour changes
thymol blue	1.2–2.8 and 8.0–9.6
naphthyl red	3.7–5.0
neutral red	6.8–8.0

Which indicators should be used to show the first and second equivalence points?

- A thymol blue for both
- B naphthyl red and then thymol blue
- C thymol blue and then neutral red
- D naphthyl red and then neutral red